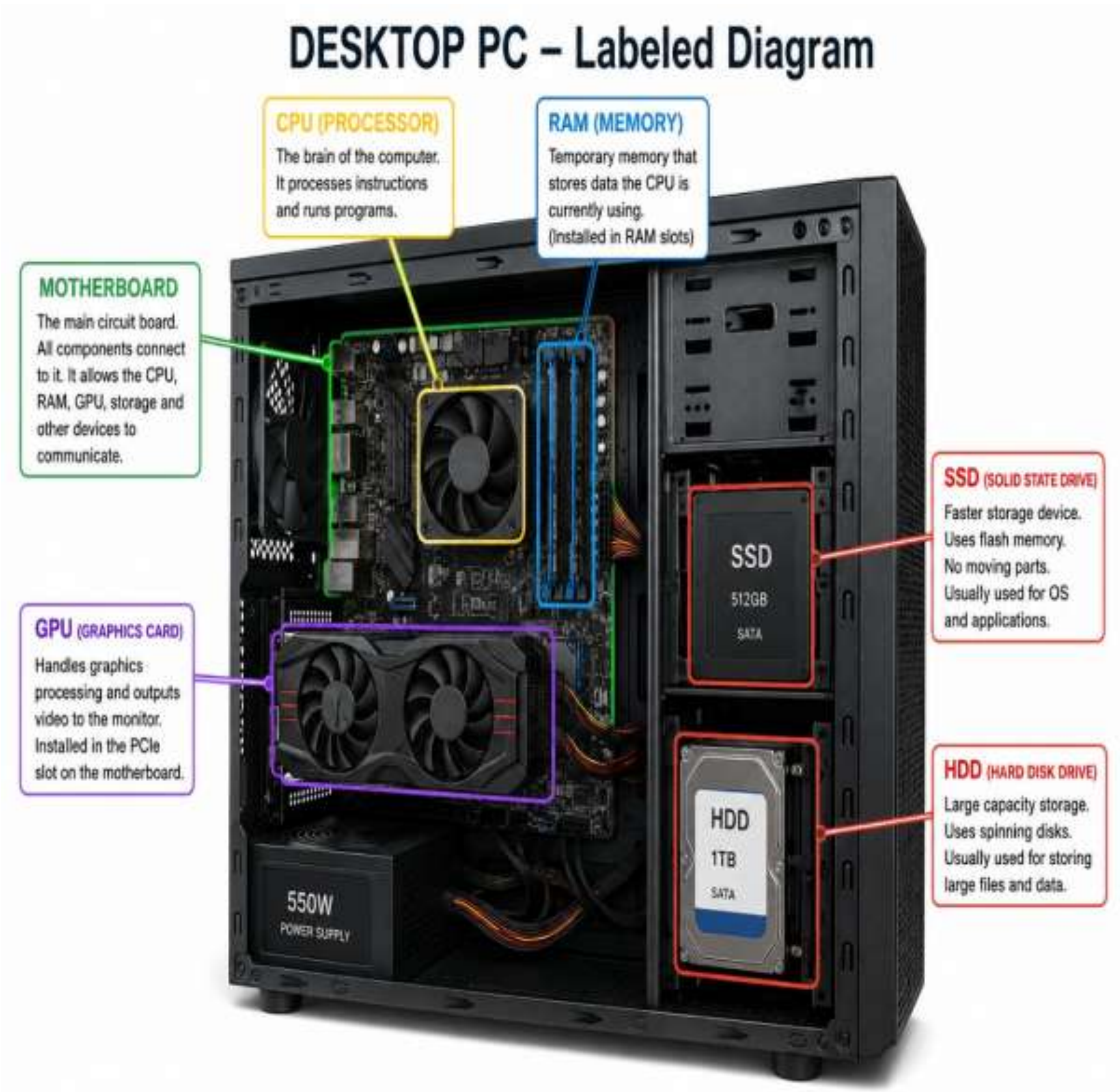
Assignment -1

- HARDWARE -

1. Draw a labelled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. **Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases. Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.**

→

Point	**HDD (Hard Disk Drive)**	**SSD (Solid State Drive**
Speed	Slow	Very Fast
Durability	Less durable (can be damaged by shock)	More durable (shock-resistant)
Moving Parts	Yes (spinning disk)	No (electronic chips)
Boot Time	Takes more time	Starts very quickly
Storage Size	Large storage for low price	Large storage for low price
Price	Cheaper	More expensive
Typical Use	Desktop PC, office computer, file storage	Laptop, Gaming PC, High-performance computer
Where It Is Used	Desktop PCs, office computers, and storage drives.	Laptops, gaming PCs, and modern computers.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

→

➢ CPU (Central Processing Unit)

- The CPU is the brain of the computer. It controls the game and tells all the parts what to do. In games like PUBG or Free Fire, the CPU handles player actions, enemies, and game rules.

➢ RAM (Random Access Memory)

- RAM is the computer's short-term memory. It keeps the game data ready while you are playing. More RAM helps the game run smoothly and reduces loading time.

➢ GPU (Graphics Processing Unit)

- The GPU is the graphics maker of the computer. It creates the game's pictures, colors, shadows, and animations. A good GPU makes PUBG or Free Fire look better and run with higher FPS (frames per second).

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Hardware	Why Upgrade It?
CPU (Processor)	A faster CPU helps the game run smoothly and handles game tasks quickly.
RAM	More RAM helps the game load faster and reduces lag while playing.
GPU (Graphics Card)	A better GPU gives better graphics, higher FPS, and smoother gameplay.
SSD	An SSD makes Windows and the game start faster. It also reduces loading time.
Power Supply (PSU)	If you install a new GPU or CPU, you may need a stronger power supply.
Cooling Fan	Better cooling keeps the PC cool and helps it work well during long gaming sessions.